

# Survival MF Derivations

Updated 3/11/26 — Notation &  $\beta$  Corrections + Algorithm

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## 1 Introduction

The following document presents derivations for matrix factorization & survival expressions for supervised matrix factorization. Content includes an expression of the overall objective function to minimize as well as the procedure for updating the prior distributions for  $L$ ,  $F$ ,  $\beta$ , and  $\tau$ . These updates are solved via the EBNM problem illustrated in the EBMF paper.

### 1A Models

The following models are considered:

- **Genomics Data:**  $Y_{n \times p} = L_{n \times K} F_{K \times p}^\top + E_{n \times p}$ , where  $E_{ij} \stackrel{\text{iid}}{\sim} N(0, \tau_j^{-1})$
- **Survival Data (Cox PH):**  $h(t_i) = h_0(t_i) \exp(\eta_i)$ , where  $\eta_i = \mathbf{l}_i^\top \beta = \sum_k l_{ik} \beta_k$

### 1B Notation: Posterior Moments

Throughout these derivations,  $q(\cdot)$  denotes the variational posterior and  $g(\cdot)$  the empirical Bayes prior. We distinguish carefully between the *posterior mean* and the *posterior second moment*:

$$\bar{l}_{ik} := E_q[l_{ik}] \quad (\text{posterior mean}) \quad (1)$$

$$\overline{l_{ik}^2} := E_q[l_{ik}^2] = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2 \quad (\text{posterior second moment}) \quad (2)$$

and analogously  $\bar{f}_{jk}$ ,  $\overline{f_{jk}^2}$ ,  $\bar{\beta}_k$ ,  $\overline{\beta_k^2}$ . These two quantities are **not** equal whenever there is posterior uncertainty, and the distinction is essential for correct EBNM inputs and for the  $\tau$  update.

### 1C Update Procedure

The general procedure for parameter updates is as follows.

**For**  $b = 1, \dots, B$ :

1. Compute survival working quantities  $\hat{\eta}_i$ ,  $u_i$ ,  $W_{ii}$ ,  $z_i$  from the current posterior means.
2. Iterate over  $k = 1, \dots, K$ :
  - (a) Update  $q(l_k)$ ,  $g(l_k)$  with other parameters fixed (different from EBMF: combines genomics & survival terms).
  - (b) Update  $q(f_k)$ ,  $g(f_k)$  with other parameters fixed (identical to the EBMF problem).
  - (c) Update  $q(\beta_k)$ ,  $g(\beta_k)$  with other parameters fixed (survival term only; solved via EBNM under mean-field independence).
3. Update  $\tau_j$  with other parameters fixed (identical to EBMF).

## 2 Objective Function

The ELBO to *maximize* over posteriors  $(q_l, q_f, q_\beta)$ , priors  $(g_l, g_f, g_\beta)$ , and precision  $\tau$  is

$$\begin{aligned} \mathcal{F}(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) = & E_{q_l, q_f, q_\beta} [\log P(Y, t, \delta \mid L, F, \beta, \tau)] \\ & + \sum_k E_{q_{l_k}} \left[ \log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + \sum_k E_{q_{f_k}} \left[ \log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + \sum_k E_{q_{\beta_k}} \left[ \log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \end{aligned} \quad (3)$$

The full data log-likelihood decomposes as

$$\log P(Y, t, \delta \mid L, F, \beta, \tau) = \underbrace{\log P(Y \mid L, F, \tau^{-1})}_{\text{Genomics}} + \underbrace{\log P(t, \delta \mid L, \beta)}_{\text{Survival}} \quad (4)$$

### 3 Taylor Series Approximation of the Survival Likelihood

Because  $\log P(t, \delta \mid L, \beta)$  is non-conjugate in  $L$  and  $\beta$ , we use a second-order Taylor expansion to obtain a tractable Gaussian surrogate.

The second-order approximation takes the form

$$f(x) = f(x_0) + \underbrace{f'(x_0)}_{\text{gradient/score}} (x - x_0) + \frac{1}{2} \underbrace{f''(x_0)}_{\text{Hessian/info}} (x - x_0)^2 \quad (5)$$

The linear predictor for the survival data is  $\eta_i = \sum_k l_{ik} \beta_k$ . Let  $\hat{\boldsymbol{\eta}}$  be the current expansion point. Define the score  $u_i$  and the negative diagonal Hessian  $W_{ii}$ :

$$u_i = \left. \frac{\partial \ell}{\partial \eta_i} \right|_{\hat{\boldsymbol{\eta}}}, \quad W_{ii} = - \left. \frac{\partial^2 \ell}{\partial \eta_i^2} \right|_{\hat{\boldsymbol{\eta}}} \geq 0 \quad (6)$$

Applying the expansion to  $\ell(\boldsymbol{\eta}) = \log P(t, \delta \mid L, \beta)$ :

$$\ell(\boldsymbol{\eta}) \approx \ell(\hat{\boldsymbol{\eta}}) + (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}})^\top \underbrace{\nabla \ell(\hat{\boldsymbol{\eta}})}_{\mathbf{u}} + \frac{1}{2} (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}})^\top \underbrace{\nabla^2 \ell(\hat{\boldsymbol{\eta}})}_{H=-W} (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}}) \quad (7)$$

$$\approx \ell(\hat{\boldsymbol{\eta}}) + \sum_i (\eta_i - \hat{\eta}_i) u_i - \frac{1}{2} \sum_i (\eta_i - \hat{\eta}_i)^2 W_{ii} \quad (8)$$

$$= \sum_i \left[ u_i \eta_i - u_i \hat{\eta}_i - \frac{1}{2} W_{ii} \eta_i^2 + W_{ii} \eta_i \hat{\eta}_i - \frac{1}{2} W_{ii} \hat{\eta}_i^2 \right] \quad (\text{keep only terms with } \eta_i) \quad (9)$$

$$= \sum_i \left[ \underbrace{\left( -\frac{1}{2} W_{ii} \right)}_A \eta_i^2 + \underbrace{(u_i + W_{ii} \hat{\eta}_i)}_B \eta_i \right] = \sum_i \left( -\frac{1}{2} W_{ii} (\eta_i - z_i)^2 \right) \quad (10)$$

where completing the square gives the **working response**:

$$z_i = \hat{\eta}_i + \frac{u_i}{W_{ii}} \quad (11)$$

Hence  $\ell(\boldsymbol{\eta}) \approx \sum_i \left[ -\frac{1}{2} W_{ii} (\eta_i - z_i)^2 \right] + C$ , which is a weighted Gaussian likelihood with response  $z_i$  and precision  $W_{ii}$ .

### 4 Update for $q_l$

To update  $q(l_k)$  and  $g(l_k)$ , we extract from  $\mathcal{F}$  the terms depending on  $l_k$ :

$$\mathcal{F}(q_l, g_l) = E_{q_l, q_f, q_\beta} [\log P(Y, t, \delta \mid L, F, \beta, \tau)] + \sum_k E_{q_{l_k}} \left[ \log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \quad (12)$$

$$= \underbrace{E[\log P(Y \mid L, F, \tau^{-1})]}_{\text{Eq. 1: Genomics}} + \underbrace{E[\log P(t, \delta \mid L, \beta)]}_{\text{Eq. 2: Survival}} + \underbrace{\sum_k E_{q_{l_k}} \left[ \log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right]}_{\text{KL terms}} \quad (13)$$

$$= \sum_k \left[ \text{Terms from } 1(l_k) + \text{Terms from } 2(l_k) + E_{q_{l_k}} \left[ \log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \right] \quad (14)$$

## 4A Partial Residual Definitions

### Item 1 & 2 Fix: Posterior-Mean Residual Definitions

The previous document defined  $R_{ij}^k$  and  $z_i^k$  without taking expectation with respect to  $q$ , and used inconsistent superscript notation ( $k$  vs.  $-k$ ). The corrected definitions below use posterior means explicitly and adopt the consistent notation  $R_{ij}^{-k}$  and  $z_i^{-k}$  throughout.

**Genomics partial residual.** The  $k$ -th residual, which removes the contributions of all factors *except*  $k$ , evaluated at posterior means, is:

$$R_{ij}^{-k} := y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'} \Rightarrow y_{ij} \approx R_{ij}^{-k} + l_{ik} f_{jk} \quad (15)$$

**Survival partial working response.** The analogous quantity for the survival term, removing all factors except  $k$ , evaluated at posterior means, is:

$$z_i^{-k} := z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \Rightarrow z_i \approx z_i^{-k} + l_{ik} \beta_k \quad (16)$$

Note:  $z_i^{-k}$  involves  $\beta$ , not  $F$ ; this corrects the previous document's inconsistency where  $f_{jk'}$  appeared in a per-sample quantity.

## 4B Genomics Likelihood Contribution (Equation 1)

$$E[\log P(Y | L, F, \tau)] = E \left[ \sum_{ij} \left[ -\frac{1}{2} \tau_j \left( y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 \right] \right] \quad (17)$$

Substituting  $y_{ij} = R_{ij}^{-k} + l_{ik} f_{jk}$ :

$$= E \left[ \sum_{ij} \left[ -\frac{1}{2} \tau_j \left( R_{ij}^{-k} - l_{ik} f_{jk} \right)^2 \right] \right] \quad (18)$$

$$= E \left[ \sum_{ij} \left[ -\frac{1}{2} \tau_j \left( \cancel{(R_{ij}^{-k})^2} - 2R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right] \right] \quad (19)$$

$(R_{ij}^{-k})^2$  is constant w.r.t.  $l_{ik}$  and drops. Taking  $E_{q_f}[\cdot]$  over  $F$ :

$$\text{Genomics A-coefficient on } l_{ik}^2 : \tau_j E_{q_f}[f_{jk}^2] = \tau_j \overline{f_{jk}^2} \quad (20)$$

$$\text{Genomics B-coefficient on } l_{ik} : \tau_j R_{ij}^{-k} E_{q_f}[f_{jk}] = \tau_j R_{ij}^{-k} \bar{f}_{jk} \quad (21)$$

## 4C Survival Likelihood Contribution (Equation 2)

Using the Taylor surrogate with  $z_i = z_i^{-k} + l_{ik} \beta_k$ :

$$E[\log P(t, \delta | L, \beta)] \approx E \left[ \sum_i \left[ -\frac{1}{2} W_{ii} \left( z_i^{-k} - l_{ik} \beta_k \right)^2 \right] \right] \quad (22)$$

$$= E \left[ \sum_i \left[ -\frac{1}{2} W_{ii} \left( \cancel{(z_i^{-k})^2} - 2z_i^{-k} l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \right] \right] \quad (23)$$

Taking  $E_{q_\beta}[\cdot]$  over  $\beta$ :

$$\text{Survival } A\text{-coefficient on } l_{ik}^2 : W_{ii} E_{q_\beta} [\beta_k^2] = W_{ii} \bar{\beta}_k^2 \quad (24)$$

$$\text{Survival } B\text{-coefficient on } l_{ik} : W_{ii} z_i^{-k} E_{q_\beta} [\beta_k] = W_{ii} z_i^{-k} \bar{\beta}_k \quad (25)$$

#### 4D Combined Coefficients and EBNM Problem

The full objective for  $q(l_k)$  takes the form

$$\begin{aligned} \mathcal{F}(q_l, g_l) = \sum_k \left[ E_{q_l, q_f, q_\beta} \left[ \sum_{ij} \left[ -\frac{1}{2} \tau_j \left( \overline{R_{ij}^{-k}}^2 - 2 R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right] \right. \right. \\ \left. \left. + \sum_i \left[ -\frac{1}{2} W_{ii} \left( \overline{z_i^{-k}}^2 - 2 z_i^{-k} l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \right] + E_{q_{l_k}} \left[ \log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \right] \right] \end{aligned} \quad (26)$$

Summing the genomics and survival contributions yields the combined coefficients:

$$\mathbf{A}_{ik} = \sum_j \tau_j \overline{f_{jk}^2} + W_{ii} \bar{\beta}_k^2 \quad (27)$$

$$\mathbf{B}_{ik} = \sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k \quad (28)$$

##### EBNM Problem

For each sample  $i$ , the update for factor  $k$  solves:

$$x_i = \frac{B_{ik}}{A_{ik}} = \frac{\sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k}{\sum_j \tau_j \overline{f_{jk}^2} + W_{ii} \bar{\beta}_k^2} \quad (29)$$

$$s_i = \frac{1}{\sqrt{A_{ik}}} \quad (30)$$

$$(\hat{g}_{l_k}, \hat{q}_{l_k}) = \text{EBNM}(\mathbf{x}, \mathbf{s}) \quad (31)$$

The EBNM solver returns  $\bar{l}_{ik} = E_{\hat{q}}[l_{ik}]$  and  $\bar{l}_{ik}^2 = \text{Var}_{\hat{q}}(l_{ik}) + \bar{l}_{ik}^2$ .

## 5 Update for $q_f$

$F$  appears only in the genomics term of the likelihood, so:

$$\mathcal{F}(q_{f_k}, g_{f_k}) \approx E_{q_l, q_f} [\log P(Y \mid L, F, \tau)] + E_{q_{f_k}} \left[ \log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] \quad (32)$$

The genomics data model is  $Y_{ij} \sim N(\sum_k l_{ik} f_{jk}, \tau_j^{-1})$ .

### 5A Defining the Partial Residual

#### Item 2 Fix: Consistent Notation for $F$ Update

The partial residual for the  $F$  update uses the same posterior-mean definition as for the  $L$  update (Eq. 15), with the consistent superscript  $-k$  notation.

The  $k$ -th genomics partial residual (same definition as in the  $L$  update, Eq. 15):

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'} \Rightarrow y_{ij} \approx R_{ij}^{-k} + l_{ik} f_{jk} \quad (33)$$

### 5B Deriving Coefficients for $f_{jk}$

Expanding the log-likelihood:

$$E_q[\log P(Y | L, F, \tau)] = \sum_{ij} E_q \left[ -\frac{1}{2} \tau_j (R_{ij}^{-k} - l_{ik} f_{jk})^2 \right] \quad (34)$$

Expanding the square and dropping the  $(R_{ij}^{-k})^2$  constant:

$$= \sum_{ij} E_q \left[ -\frac{1}{2} \tau_j \left( \cancel{(R_{ij}^{-k})^2} - 2R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right] \quad (35)$$

Taking  $E_q[\cdot]$  over  $L$ , treating  $R_{ij}^{-k}$  as fixed, and grouping over  $i$  (for each fixed  $j$ ) to read off the quadratic form in  $f_{jk}$ :

$$\mathcal{F}(q_{f_k}) \approx E_q \left[ \sum_j \left[ -\frac{1}{2} f_{jk}^2 \underbrace{\left( \sum_i \tau_j \bar{l}_{ik}^2 \right)}_{A_{jk}} + f_{jk} \underbrace{\left( \sum_i \tau_j R_{ij}^{-k} \bar{l}_{ik} \right)}_{B_{jk}} \right] \right] + E_{q_{f_k}} \left[ \log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] \quad (36)$$

The coefficients are:

$$\mathbf{A}_{jk} = \sum_i \tau_j \bar{l}_{ik}^2 \quad (37)$$

$$\mathbf{B}_{jk} = \sum_i \tau_j R_{ij}^{-k} \bar{l}_{ik} \quad (38)$$

#### EBNM Problem

For each feature  $j$ , solve the EBNM problem (subscript  $j$ , not  $i$ ):

$$x_j = \frac{B_{jk}}{A_{jk}} = \frac{\sum_i \tau_j R_{ij}^{-k} \bar{l}_{ik}}{\sum_i \tau_j \bar{l}_{ik}^2} \quad (39)$$

$$s_j = \frac{1}{\sqrt{A_{jk}}} \quad (40)$$

$$(\hat{g}_{f_k}, \hat{q}_{f_k}) = \text{EBNM}(\mathbf{x}, \mathbf{s}) \quad (41)$$

## 6 Update for $q_\beta$

### Item 3 Fix: Removed WLS Derivation; EBNM Approach Only

The previous document contained two approaches for updating  $\beta$ : a weighted least-squares (WLS) formulation (Sec. 6) and an EBNM formulation (Sec. 6A). The key difference is the *posterior independence assumption*: Sec. 6A assumes  $q(\beta) = \prod_k q(\beta_k)$ , so that  $E_q[\beta_k \beta_{k'}] = E_q[\beta_k] E_q[\beta_{k'}]$  for  $k \neq k'$ . This assumption greatly simplifies the math and enables a coordinate-ascent EBNM update that is consistent with the updates for  $L$  and  $F$ . The old Sec. 6 WLS derivation has been **removed** to avoid confusion.

## 6A Setting Up the Objective

$\beta$  appears only in the survival term. The objective for  $q_\beta$  is:

$$\mathcal{F}(q_\beta, g_\beta) = E_{q_l, q_\beta}[\log P(t, \delta \mid L, \beta)] + \sum_k E_{q_{\beta_k}} \left[ \log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (42)$$

Using the Taylor approximation:

$$\mathcal{F}(q_\beta, g_\beta) \approx E_{q_l, q_\beta} \left[ -\frac{1}{2} \sum_i W_{ii} \left( z_i - \sum_k l_{ik} \beta_k \right)^2 \right] + \sum_k E_{q_{\beta_k}} \left[ \log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (43)$$

## 6B Variance-Mean Decomposition

A key identity:

$$E_{q_l} [l_{ik}^2] = \text{Var}_{q_l}(l_{ik}) + (E_{q_l}[l_{ik}])^2 \Rightarrow \bar{l}_{ik}^2 = \text{Var}_{q_l}(l_{ik}) + \bar{l}_{ik}^2 \quad (44)$$

## 6C Coordinate-Ascent Update for $\beta_k$

By applying the mean-field independence assumption  $q(\beta) = \prod_k q(\beta_k)$ , we update a single  $\beta_k$  while fixing all  $\beta_{k'}$  for  $k' \neq k$ .

The survival partial working response (consistent with Eq. 16):

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \Rightarrow z_i \approx z_i^{-k} + l_{ik} \beta_k \quad (45)$$

Substituting into Eq. (43):

$$-\frac{1}{2} \sum_i W_{ii} (z_i^{-k} - l_{ik} \beta_k)^2 = \sum_i -\frac{1}{2} W_{ii} \left( \cancel{(z_i^{-k})^2} - 2z_i^{-k} l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \quad (46)$$

Taking  $E_{q_l}[\cdot]$ , using  $E_{q_l}[l_{ik}] = \bar{l}_{ik}$  and  $E_{q_l}[l_{ik}^2] = \bar{l}_{ik}^2$ :

$$\mathcal{F}(q_{\beta_k}) = \sum_i \left[ W_{ii} z_i^{-k} \bar{l}_{ik} \beta_k - \frac{1}{2} W_{ii} \bar{l}_{ik}^2 \beta_k^2 \right] + E_{q_{\beta_k}} \left[ \log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] + C \quad (47)$$

## 6D Formulating the EBNM Problem

### Item 4 Fix: Remove Erroneous $1/\sigma^2$ Term from $A_k$

The previous document (Sec. 6A3, Eq. 91) included  $+1/\sigma^2$  in  $A_k$ , accounting for the prior  $g(\beta_k) \sim N(0, \sigma^2)$  explicitly. However, as stated in Sec. 6A, the goal is to reduce the  $\beta_k$  update to a *1D EBNM problem* in which the EBNM solver empirically fits  $g_{\beta_k}$  directly from the data. Adding  $1/\sigma^2$  to  $A_k$  therefore accounts for the prior **twice**—once through the EBNM prior estimation and again through the explicit term. The corrected  $A_k$  below omits the  $1/\sigma^2$  term. Correspondingly,  $x_k$  is updated by removing the  $1/\sigma^2$  from the denominator. Additionally, the posterior mean  $\bar{l}_{ik}$  (not just  $l_{ik}$ ) is used in  $B_k$  for consistency.

Grouping Eq. (47) into the standard quadratic form  $-\frac{1}{2} A_k \beta_k^2 + B_k \beta_k$ :

$$\mathbf{A}_k = \sum_i W_{ii} \bar{l}_{ik}^2 \quad (48)$$

$$\mathbf{B}_k = \sum_i W_{ii} z_i^{-k} \bar{l}_{ik} \quad (49)$$

Using  $\bar{l}_{ik}^2$  (full second moment) rather than  $\bar{l}_{ik}^2$  in  $A_k$  encodes an *error-in-variables correction*: posterior uncertainty in  $L$  appropriately inflates the effective noise for  $\beta_k$ , preventing overfitting.

#### EBNM Problem

For each factor  $k$ , solve the 1D EBNM problem with a single pseudo-observation:

$$x_k = \frac{B_k}{A_k} = \frac{\sum_i W_{ii} z_i^{-k} \bar{l}_{ik}}{\sum_i W_{ii} \bar{l}_{ik}^2} \quad (50)$$

$$s_k = \frac{1}{\sqrt{A_k}} \quad (51)$$

$$(\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k) \quad (52)$$

**Remark.** The EBNM solver empirically fits  $g_{\beta_k}$  from the data, so no fixed  $\sigma^2$  is needed in  $A_k$ . The prior regularisation is entirely handled by the EBNM estimation step.

## 7 Update for $\tau$

#### Item 5: No Changes (Confirmed Correct from EBMF)

This section is unchanged from the previous document. It follows section A.1 of the EBMF paper (Equations A.5–A.11) exactly and has been confirmed correct.

We derive updates to optimize  $\mathcal{F}$  over the precision parameters  $\tau$  identically to section A.1 of the EBMF paper. We focus on the components of  $\mathcal{F}$  that depend on  $\tau$ :

$$\mathcal{F}(\tau) = E_{q_l} E_{q_f} \sum_{ij} \frac{1}{2} \log(\tau_j) - \frac{1}{2} \tau_j \left( Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 + C \quad (53)$$

$$= \frac{1}{2} \sum_{ij} \left[ \log(\tau_j) - \tau_j \bar{R}_{ij}^2 \right] + C \quad (54)$$

where  $\bar{R}_{ij}^2$  is defined by:

$$\bar{R}_{ij}^2 := E_{q_l, q_f} \left[ \left( Y_{ij} - \sum_{k=1}^K l_{ik} f_{jk} \right)^2 \right] \quad (55)$$

$$= \left( Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2 - \sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2 + \sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2 \quad (56)$$

If  $\tau \in \mathcal{T}$  is constrained:

$$\hat{\tau} = \arg \max_{\tau \in \mathcal{T}} \sum_{ij} \left[ \log(\tau_j) - \tau_j \bar{R}_{ij}^2 \right] \quad (57)$$

Assuming column-specific precisions  $\tau_{ij} = \tau_j$ :

$$\hat{\tau}_j = \frac{N}{\sum_i \bar{R}_{ij}^2} \quad (58)$$



Assuming constant precision  $\tau_{ij} = \tau$ :

$$\hat{\tau} = \frac{NP}{\sum_{ij} \overline{R^2}_{ij}} \quad (59)$$

In Equations 58 and 59,  $N$  is the number of samples (rows of  $Y$ ) and  $P$  is the number of features (columns of  $Y$ ).

## 8 Algorithm

The following algorithm presents the complete coordinate-ascent variational inference (CAVI) procedure for the supervised matrix factorization model. It follows the style of Algorithms 3 and 5 in the EBMF paper, with additional detail to facilitate implementation and debugging. Equation numbers refer to sections above.

### Algorithm 1. Supervised MF — CAVI Update Procedure

#### Require:

- Genomics data matrix  $Y_{n \times p}$
- Survival data  $(t_i, \delta_i)$  for  $i = 1, \dots, n$
- Number of latent factors  $K$ ; convergence tolerance  $\varepsilon$

#### Initialize ( $b = 0$ ):

- Set  $\bar{l}_{ik}^{(0)}, \bar{f}_{jk}^{(0)}$  via rank- $K$  SVD of  $Y$
- Set  $\overline{l_{ik}^2}^{(0)} = \bar{l}_{ik}^{(0)^2}, \overline{f_{jk}^2}^{(0)} = \bar{f}_{jk}^{(0)^2}$  (initialise second moments at squared means)
- Set  $\bar{\beta}_k^{(0)}$  via a Cox PH fit on the initial loadings  $\bar{L}^{(0)}$ ; set  $\overline{\beta_k^2}^{(0)} = \bar{\beta}_k^{(0)^2}$
- Set  $\hat{\tau}_j^{(0)}$  via column-wise sample variance of  $Y$

#### For $b = 1, 2, \dots$ until convergence:

**Step 1 — Compute Survival Working Quantities** (once per outer iteration, using posterior means from iteration  $b - 1$ )

1.1. Compute the current linear predictor for each sample  $i$ :

$$\hat{\eta}_i^{(b)} = \sum_{k=1}^K \bar{l}_{ik}^{(b-1)} \bar{\beta}_k^{(b-1)} \quad (\text{using posterior means from iteration } b-1)$$

1.2. Evaluate the Cox partial likelihood score and negative Hessian at  $\hat{\eta}^{(b)}$ :

$$u_i^{(b)} = \left. \frac{\partial \ell_{\text{Cox}}}{\partial \eta_i} \right|_{\hat{\eta}_i^{(b)}}, \quad W_{ii}^{(b)} = - \left. \frac{\partial^2 \ell_{\text{Cox}}}{\partial \eta_i^2} \right|_{\hat{\eta}_i^{(b)}} \geq 0$$

1.3. Compute the working response  $z_i$  (Sec. 3):

$$z_i^{(b)} = \hat{\eta}_i^{(b)} + \frac{u_i^{(b)}}{W_{ii}^{(b)}}$$

#### Step 2 — Update Factors (for $k = 1, \dots, K$ )

For each  $k$ , perform sub-steps (a), (b), (c) in sequence:

(a) **Update**  $q(l_k), g(l_k)$  — Patient Loadings (combines genomics & survival; Eqs. 27–30)

**a.1.** Compute genomics partial residual using current posterior means:

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'}^{(b)} \bar{f}_{jk'}^{(b)}$$

(Use the *most recently updated* means for  $k' < k$  in iteration  $b$ , and means from iteration  $b - 1$  for  $k' > k$ .)

**a.2.** Compute survival partial working response:

$$z_i^{-k} = z_i^{(b)} - \sum_{k' \neq k} \bar{l}_{ik'}^{(b)} \bar{\beta}_{k'}^{(b)}$$

**a.3.** Compute the EBNM precision and sufficient statistic (Eqs. 27–28):

$$\begin{aligned} A_{ik} &= \sum_j \hat{\tau}_j^{(b-1)} \overline{f_{jk}^2}^{(b-1)} + W_{ii}^{(b)} \overline{\beta_k^2}^{(b-1)} \\ B_{ik} &= \sum_j \hat{\tau}_j^{(b-1)} R_{ij}^{-k} \bar{f}_{jk}^{(b-1)} + W_{ii}^{(b)} z_i^{-k} \bar{\beta}_k^{(b-1)} \end{aligned}$$

**a.4.** Form the EBNM pseudo-observation and noise:

$$x_i = \frac{B_{ik}}{A_{ik}}, \quad s_i = \frac{1}{\sqrt{A_{ik}}}$$

**a.5.** Solve the EBNM problem (Eq. 29):

$$\left( \hat{g}_{l_k}^{(b)}, \hat{q}_{l_k}^{(b)} \right) = \text{EBNM}(\{x_i\}_{i=1}^n, \{s_i\}_{i=1}^n)$$

Store  $\bar{l}_{ik}^{(b)} = E_{\hat{q}}[l_{ik}]$  and  $\overline{l_{ik}^2}^{(b)} = \text{Var}_{\hat{q}}(l_{ik}) + \bar{l}_{ik}^{(b)2}$ .

**(b) Update  $q(f_k), g(f_k)$  — Biological Factors** (*genomics only; identical to EBMF; Eqs. 37–40*)

**b.1.** Recompute the genomics partial residual using *updated*  $\bar{l}_{ik}^{(b)}$ :

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'}^{(b)} \bar{f}_{jk'}^{(b)}$$

**b.2.** Compute the EBNM precision and sufficient statistic (Eqs. 37–38):

$$\begin{aligned} A_{jk} &= \sum_i \hat{\tau}_j^{(b-1)} \overline{l_{ik}^2}^{(b)} \\ B_{jk} &= \sum_i \hat{\tau}_j^{(b-1)} R_{ij}^{-k} \bar{l}_{ik}^{(b)} \end{aligned}$$

**b.3.** Form the EBNM pseudo-observation and noise:

$$x_j = \frac{B_{jk}}{A_{jk}}, \quad s_j = \frac{1}{\sqrt{A_{jk}}}$$

**b.4.** Solve the EBNM problem (Eq. 39):

$$\left( \hat{g}_{f_k}^{(b)}, \hat{q}_{f_k}^{(b)} \right) = \text{EBNM}(\{x_j\}_{j=1}^p, \{s_j\}_{j=1}^p)$$

Store  $\bar{f}_{jk}^{(b)} = E_{\hat{q}}[f_{jk}]$  and  $\overline{f_{jk}^2}^{(b)} = \text{Var}_{\hat{q}}(f_{jk}) + \bar{f}_{jk}^{(b)2}$ .

**(c) Update**  $q(\beta_k), g(\beta_k)$  — Survival Coefficients *(survival only; 1D EBNM; Eqs. 48–51)*

**c.1.** Compute the survival partial working response using updated  $\bar{l}_{ik}^{(b)}$  and  $\bar{\beta}_k^{(b)}$  (latest available):

$$z_i^{-k} = z_i^{(b)} - \sum_{k' \neq k} \bar{l}_{ik'}^{(b)} \bar{\beta}_{k'}^{(b)}$$

**c.2.** Compute the EBNM precision and sufficient statistic (Eqs. 48–49):

$$A_k = \sum_i W_{ii}^{(b)} \bar{l}_{ik}^{2(b)}$$

$$B_k = \sum_i W_{ii}^{(b)} z_i^{-k} \bar{l}_{ik}^{(b)}$$

*Note: No  $1/\sigma^2$  term is added to  $A_k$  — the prior is handled internally by the EBNM solver.*

**c.3.** Form the EBNM pseudo-observation and noise:

$$x_k = \frac{B_k}{A_k}, \quad s_k = \frac{1}{\sqrt{A_k}}$$

**c.4.** Solve the 1D EBNM problem (Eq. 50):

$$\left( \hat{g}_{\beta_k}^{(b)}, \hat{q}_{\beta_k}^{(b)} \right) = \text{EBNM}(x_k, s_k)$$

Store  $\bar{\beta}_k^{(b)} = E_{\hat{q}}[\beta_k]$  and  $\bar{\beta}_k^{2(b)} = \text{Var}_{\hat{q}}(\beta_k) + \bar{\beta}_k^{(b)2}$ .

**end for** (over  $k = 1, \dots, K$ )

**Step 3 — Update**  $\hat{\tau}$  — Noise Precision *(identical to EBMF; Eq. 58)*

**3.1.** Compute the expected squared residual for each entry  $(i, j)$ :

$$\bar{R}_{ij}^2 = \left( Y_{ij} - \sum_k \bar{l}_{ik}^{(b)} \bar{f}_{jk}^{(b)} \right)^2 + \sum_k \left[ \bar{l}_{ik}^{2(b)} \bar{f}_{jk}^{2(b)} - \bar{l}_{ik}^{(b)2} \bar{f}_{jk}^{(b)2} \right]$$

**3.2.** Update precision (column-specific):

$$\hat{\tau}_j^{(b)} = \frac{N}{\sum_i \bar{R}_{ij}^2}$$

**Step 4 — Check Convergence**

Compute the maximum absolute change in loadings and survival coefficients:

$$\Delta_L^{(b)} = \max_{i,k} \left| \bar{l}_{ik}^{(b)} - \bar{l}_{ik}^{(b-1)} \right|, \quad \Delta_\beta^{(b)} = \max_k \left| \bar{\beta}_k^{(b)} - \bar{\beta}_k^{(b-1)} \right|$$

**If**  $\max(\Delta_L^{(b)}, \Delta_\beta^{(b)}) < \varepsilon$ : stop and declare convergence.

**Else:** continue to iteration  $b + 1$ .

**Return:** Posterior means  $\bar{L} = \{\bar{l}_{ik}\}$ ,  $\bar{F} = \{\bar{f}_{jk}\}$ ,  $\bar{\beta} = \{\bar{\beta}_k\}$ , and posterior second moments  $\bar{L}^2$ ,  $\bar{F}^2$ ,  $\bar{\beta}^2$ , and noise precision  $\hat{\tau}$ .

## Notes on the Algorithm

**Working quantities ( $z_i, W_{ii}$ ) computed once per outer iteration.** The quantities  $z_i^{(b)}$  and  $W_{ii}^{(b)}$  are computed once at the start of each outer loop (Step 1), using the linear predictor from iteration  $b-1$ . Within the inner  $k$ -loop, as  $\bar{L}$  and  $\bar{\beta}$  are updated, the true linear predictor changes but  $z_i$  and  $W_{ii}$  are not refreshed. This is a standard IRLS-within-VI approximation; the outer loop restores accuracy at each iteration  $b$ .

**Single-observation EBNM for  $\beta_k$ .** For each  $k$ , the  $\beta_k$  EBNM call uses a single pseudo-observation  $x_k$  aggregated over all samples  $i$ . The prior  $g_{\beta_k}$  is estimated from this one point, which may be highly variable for small  $K$ .

**Second moments and posterior variance.** The algorithm uses  $\overline{l_{ik}^2} = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2$  (not just  $\bar{l}_{ik}^2$ ) throughout. This *error-in-variables correction* inflates the effective noise for  $\beta_k$  in proportion to the posterior uncertainty in  $L$ , preventing overfitting.